

- Built execution patterns that routed Prefect workloads to EC2 or Fargate based on runtime conditions such as file size and processing requirements.
- Created EC2- and Fargate-backed microservices for OCR, document conversion, data extraction, and backend processing workflows.
- Designed and maintained REST APIs and message queue architecture for high-load document-processing services, improving resilience and throughput in production data exchange.
- Automated build, test, and deployment workflows for Python and JavaScript services using GitHub Actions and AWS CodePipeline.

Data Engineer

2019–2022

Oak Ridge National Laboratory

Oak Ridge, TN

- Replaced a brittle cron-based web scraping process with a more reliable Airflow ETL workflow deployed in Kubernetes and cloud infrastructure.
- Developed full-stack Docker images and automated spatial-temporal data acquisition workflows using Python, R, Airflow, and Shiny for WSTAMP and Hadoop-backed data storage.
- Led development for a COVID-19 tracking project spanning R, Python, geospatial analysis, downstream model creation, and operational reporting workflows; received distinguished recognition for the project.
- Built containerized Airflow ETL workflows, Python microservices, and Swagger/OpenAPI REST APIs for NAERM-RTSA and related data exchange with PostgreSQL, AWS S3, AWS RDS, and MongoDB.
- Automated web scraping and social media ingestion for misinformation classification, statistical modeling, network analysis, Elasticsearch indexing, and Kibana visualization.
- Served as Software Quality Assurance board chair for the Geospatial Science and Human Security Division and contributed to ORNL Data Council discussions on data management, governance, and architecture.

Data Scientist II Subcontractor

2018–2019

Oak Ridge National Laboratory

Oak Ridge, TN

- Automated WSTAMP web application data ingestion using R, Python, Airflow, and Docker.
- Authored an R package to normalize geographic identifiers and unify data access across online APIs.
- Developed a SpatiaLite-backed SQLite database to simplify lookup, retrieval, and integration of geographic data.
- Built a Python zonal statistics application for calculating raster summaries across shapefile geometries.

Owner & Contractor-For-Hire

2018

Specrabella

Knoxville, TN

- Designed and deployed interactive energy informatics applications using Leaflet, Shiny, Snowflake, and machine-learning-derived calculations.
- Consulted on greenfield web application architecture, database design, visualization, and deployment for energy-sector clients.
- Built data ingestion and visualization workflows using cURL, PHP, R, Python, and Snowflake for energy datasets.
- Designed PostgreSQL databases and development workflows for energy and financial data products.

Bioinformatician, Project Manager, & Technical Sales Representative

2016–2018

Microbial Insights

Knoxville, TN

- Established an automated NGS analysis pipeline using R, MySQL, Python, Bash, JavaScript, and Shiny-backed internal tooling.
- Automated client-facing statistical workflows including linear models, ANOVA, self-organizing maps, clustering, PCA, and PCoA.
- Developed ETL and customer data visualization tooling using R, PHP, jQuery, and MySQL for dynamic qPCR result analysis.
- Built a custom R package, local Shiny applications, and an internal PostgreSQL/PHP wiki to improve reporting, project management, and customer service workflows.

Graduate Research Assistant

2014–2016

Dr. Neal Stewart's Plant Biotechnology Lab, University of Tennessee

Knoxville, TN

- Analyzed cell suspension experiments using linear models, ANOVA, PCA, Python, and R for DOE reporting and scientific communication.
- Evaluated suspension cultures through chemical and spectral processes and translated findings into research deliverables.
- Collaborated on a novel single-cell suspension and cryopreservation robotic system while executing monocot genetic modification workflows.

Laboratory Assistant

2012–2014

Dr. Neal Stewart's Plant Biotechnology Lab, University of Tennessee

Knoxville, TN

- Developed Python-based statistical methodology for screening lignin content and imaging cell characteristics both *in vivo* and *in vitro*.

- Extracted genomes from NCBI, cleaned and normalized datasets, and performed exploratory data analysis using Python.

Laboratory Assistant

2002–2012

Dr. Paris Lambdin's Biosystematics and Biological Control Lab, University of Tennessee

Knoxville, TN

- Designed instructional modules for undergraduate and graduate courses using HTML, CSS, JavaScript, and Flash.
- Performed field data collection and forest coverage analysis using ArcGIS in support of biological control and biosystematics projects.

Selected Projects

Where I've Been — Travel history visualization application with spatial data storage, geospatial retrieval, and web mapping.

Technologies: Python, Flask, Docker, PostgreSQL, Leaflet, Vue, Bootstrap

Final Fantasy Football — Predictive analytics and semantic learning application for fantasy football data exploration and decision support.

Technologies: Python, Scikit-learn, Pandas, NumPy, Flask, Docker, Airflow

Decentralized Content Reward System — Prototype web platform exploring decentralized content incentives, workflow design, and full-stack application architecture.

Technologies: Python, Rust, Flask, Docker, PostgreSQL, React, Bootstrap

Education

Master of Science in Plant Sciences – Plant Molecular Genetics

University of Tennessee

Spring 2017

GPA: 3.72/4.0

Bachelor of Science in Plant Sciences – Biotechnology

University of Tennessee

Spring 2014

GPA: 3.74/4.0; Magna Cum Laude

Selected Publications, Technical Reports, & Research Outputs

- [1] James D. Gaboardi et al. "Scalable Generation of High-fidelity Synthetic Population Ensembles". In: *Future Generation Computer Systems* 185 (2026), p. 108619. ISSN: 0167-739X. DOI: <https://doi.org/10.1016/j.future.2026.108619>. URL: <https://www.sciencedirect.com/science/article/pii/S0167739X26002530>.
- [2] Heidi A. Hanson et al. *Centralized Health and Exposomic Resource (C-HER): Analytic and AI-Ready Data for External Exposomic Research*. 2025. arXiv: 2511.03750 [stat.AP]. URL: <https://arxiv.org/abs/2511.03750>.
- [3] Hilda B. Klasky et al. *VA EDH Advanced Software Pipeline Framework Report: Enhancing Automation and Scalability*. Tech. rep. Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States), Mar. 2025. DOI: 10.2172/2573480. URL: <https://www.osti.gov/biblio/2573480>.
- [4] Greg Simpson et al. *Geospatial Data Workflow Orchestration and Architecture*. Tech. rep. Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States), July 2025. DOI: 10.2172/3362865. URL: <https://www.osti.gov/biblio/3362865>.
- [5] Robert Stewart et al. "Secure Route: Roadway Risk Mapping for Transportation Planners". In: Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States). Aug. 2025. URL: <https://www.osti.gov/biblio/3368818>.
- [6] Christa Brelsford et al. "A dataset of recorded electricity outages by United States county 2014–2022". In: *Scientific Data* 11.1 (2024), p. 271.
- [7] Alina Peluso et al. "Environmental determinants of health: Measuring multiple physical environmental exposures at the United States census tract level". In: *Health & Place* 89 (2024), p. 103303. ISSN: 1353-8292. DOI: <https://doi.org/10.1016/j.healthplace.2024.103303>. URL: <https://www.sciencedirect.com/science/article/pii/S135382922400131X>.
- [8] Varisara Tansakul et al. *EAGLE-I Power Outage Data 2014-2022*. Tech. rep. Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States). Oak Ridge ..., 2023.
- [9] Varisara Tansakul et al. *EAGLE-I Power Outage Data Information*. Tech. rep. Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States). Oak Ridge ..., 2023.

- [10] Robert Stewart et al. "Near real time monitoring and forecasting for COVID-19 situational awareness". In: *Applied Geography* 146 (2022), p. 102759. ISSN: 0143-6228. DOI: <https://doi.org/10.1016/j.apgeog.2022.102759>. URL: <https://www.sciencedirect.com/science/article/pii/S0143622822001308>.
- [11] Dasha Herrmannova et al. "Challenges in Automated Detection of COVID-19 Misinformation". In: (May 2021). URL: <https://www.osti.gov/biblio/1783004>.
- [12] Robert Stewart et al. "A Taxonomic Classification Approach for Global Spatio-temporal Data". In: (2021).
- [13] Joshua N Grant et al. "Improved tissue culture conditions for the emerging C 4 model *Panicum hallii*". In: *BMC biotechnology* 17.1 (2017), p. 39.
- [14] Joshua Nathaniel Grant. "Evaluation of Hall's Panicgrass (*Panicum hallii* Vasey) as a Model System for Genetic Modification of Recalcitrance in Switchgrass (*Panicum virgatum* (L.))" MA thesis. The University of Tennessee, Knoxville, 2017.
- [15] Jonathan D Willis et al. "The TcEG1 beetle (*Tribolium castaneum*) cellulase produced in transgenic switchgrass is active at alkaline pH and auto-hydrolyzes biomass for increased cellobiose release". In: *Biotechnology for Biofuels* 10.1 (2017), p. 230.
- [16] Jinyi Liu et al. "Soybean kinome: functional classification and gene expression patterns". In: *Journal of experimental botany* 66.7 (2015), pp. 1919–1934.
- [17] Joshua N Grant, Jonathan D Willis, and C Neal Stewart. "Comparisons of Switchgrass Cell Wall Components Containing an Overexpression of a Putative Switchgrass Endoglucanase". In: *IN VITRO CELLULAR & DEVELOPMENTAL BIOLOGY-PLANT*. Vol. 49. 4. SPRINGER 233 SPRING ST, NEW YORK, NY 10013 USA. 2013, pp. 475–475.
- [18] H. L. Baxter et al. "Field-based experiments on low-lignin switchgrass as a feedstock for lignocellulosic biofuel production". In: 2012.
- [19] Joshua N. Grant, Jonathan D. Willis, and C. Neal Stewart. "Functional analysis of a putative membrane-bound switchgrass endo- β -1,4-glucanase in *Escherichia coli*." In: 2012.